

Inventory Stock Analysis

What the numbers are telling us about stock, money, and gaps

Introduction

This report looks at inventory across four types of shops: agrovets, mini supermarkets, pharmacies, and electronics stores, using about two years of stock records. The goal is simple. Show where money is slipping away, show which products keep running out, and suggest what to do about both. Every number below comes from the data collected. Nothing is guessed.

The Headline Numbers

KES 48.8M Total revenue	KES 4.36M Lost to shrinkage	8.9% Shrinkage vs revenue	2,918 Stockout risk events
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Insights

Picture the business as a bucket that money pours into. Sales fill it up, but there is a small hole near the bottom. Stock that goes missing, gets damaged, or simply disappears leaks out before it can ever be counted as profit. That leak is called shrinkage, and right now about 9 shillings out of every 100 earned leak out before reaching the till. With an average profit margin of 30.1 percent, that leak takes a real bite out of hard-earned profit.

Agrovet shops lose the most money to this leak, KES 3.47 million, even though mini supermarkets report shrinkage happening more often. Agrovets don't spring a leak as frequently, but when they do, it costs far more. Fertilizer alone accounts for KES 2.997 million of the total loss, close to 69 percent of everything lost, led by DAP Fertilizer 50kg and CAN Fertilizer 50kg.

Running out of stock is a different problem, but it hurts just as much. Agrovets are again the most exposed, at risk of running out on 72 percent of days, followed by mini supermarkets at just over 50 percent. Electronics shops are safest at 32 percent.

Expensive products are not the ones leaking the most. The link between profit margin and shrinkage is weak, so the leak is more about which items get handled and moved the most, not which are worth the most.

The clearest warning sign is speed. Products whose sales are climbing fast run out far more often, 64 percent of their days carry stockout risk, compared with 39 percent for products whose sales are falling. Fertilizer makes this worse since it also takes the longest time to arrive from suppliers, 7 days on average, like ordering something urgent and being told it will show up next week. Selling fast while restocking slowly is a recipe for running dry unless a bigger safety cushion is kept.

Identified Gaps

- About 80 percent of supplier records are Nairobi-based, so it makes sense that most shrinkage value (93 percent) trace back there too. But Nairobi's shrinkage share is still higher than its share of supplier records, while Kiambu's is lower (10 percent of records, only 2 percent of shrinkage). So, beyond the volume effect, Nairobi still runs somewhat leakier per record. [uncertain why]
- There is no record of why stock goes missing, whether it is theft, damage, or spoilage. The data shows how much and where, not why.
- This dataset was built to reflect realistic patterns rather than pulled directly from live business records, so treat exact figures as directional rather than exact real-world amounts.
- There is no direct profit-per-product figure, only category-level margin. So "best performing product" below is judged by how safely stocked a product is, not by how much profit it earns.

Best Stocked vs. Most At-Risk Products

Looking at how many days of stock, each product has left right now, two clear groups stand out.

Product	Category	Stock position
Infinix Charger 33W	Electronics	Healthy, 7 to 8 days of stock left

Product	Category	Stock position
Earphones Generic	Electronics	Healthy, 7 to 8 days of stock left
DAP Fertilizer 50kg	Fertilizers	Critical, under 2 days of stock left
Elianto Cooking Oil 1L	Oils & Fats	Critical, under 2 days of stock left
Omo Washing Powder 1kg	Detergents	Critical, under 2 days of stock left
Pembe Unga 1kg	Flour & Grains	Critical, under 2 days of stock left
Salit Sugar 1kg	Sugar & Spices	Critical, under 2 days of stock left

Recommendations

- Put fertilizer first in line for closer management. It causes the biggest money loss and takes the longest to restock, so fixing it protects the most profit.
- Order fertilizer earlier than feels natural. Since it takes 7 days to arrive, place the order as soon as stock drops below the safety cushion, not when the shelf is nearly bare.
- Check fast-growing products more often than slow ones. Rising demand is the strongest early warning sign of an upcoming stockout.
- Look closely at how agrovet shops handle stock day to day, since this business type carries both the highest loss and the highest stockout risk at the same time.
- Where practical, explore suppliers outside Nairobi to reduce reliance on one region. [uncertain, since we do not know if alternative suppliers exist for every product]
- Use the reorder numbers already calculated as a starting point for the next order, for example about 166 bags of DAP Fertilizer and 173 bags of CAN Fertilizer.

Conclusion

Fertilizer is where this business is losing the most money and running out the fastest, so fixing how it is ordered and handled first will protect more profit than any other single change.